

GRAY AFF, WA, USA

WMO: 742071

Lat: 47.083N

Lon: 122.583W

Elev: 300

StdP: 14.54

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24201

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.9	24.8	13.6	11.1	27.7	19.0	14.4	28.4	23.2	49.5	20.1	48.1	3.4	140	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	87.8	66.1	83.7	64.8	80.5	63.4	68.0	83.1	66.2	80.1	64.7	77.1	7.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	87.6	70.0	61.4	82.1	68.7	59.9	78.0	68.0	32.2	83.1	30.9	80.3	29.7	77.2	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	16.1	13.4	DB	15.7	94.0	4.8	3.7	12.2	96.6	9.4	98.8	6.6	100.9	3.1	103.5	
			WB	14.8	70.4	5.3	1.9	11.0	71.7	7.8	72.8	4.9	73.9	1.0	75.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	40.3	41.3	44.9	48.8	55.1	60.0	65.6	65.7	60.3	51.3	44.2	38.9
	DBStd	10.84	6.12	5.54	4.94	4.93	5.51	5.14	4.79	4.27	4.74	4.96	6.34	6.85
	HDD50	1401	305	246	172	79	14	0	0	0	1	43	192	349
	HDD65	5118	767	663	623	486	310	167	49	43	151	425	624	810
	CDD50	1919	3	2	14	43	172	301	483	485	311	83	18	4
	CDD65	164	0	0	0	0	3	18	67	64	12	0	0	0
	CDH74	2596	0	0	1	19	124	332	972	867	277	4	0	0
	CDH80	812	0	0	0	3	24	111	344	271	59	0	0	0
(14) Wind	WSAvg	5.8	5.6	5.8	6.6	6.4	6.0	6.2	5.9	5.6	5.3	5.5	5.8	5.5
(15) Precipitation	PrecAvg	45.2	6.5	4.6	4.9	3.3	2.2	1.7	0.6	1.1	1.7	4.4	7.1	6.6
	PrecMax	56.7	13.6	11.0	9.5	8.0	3.9	3.6	1.7	3.7	8.3	10.9	16.1	12.6
	PrecMin	33.5	2.8	0.1	1.4	1.1	0.1	0.2	0.1	0.0	0.0	0.6	1.7	1.7
	PrecStd	7.3	2.4	2.3	1.9	1.8	1.1	0.9	0.5	0.9	1.8	2.2	3.4	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	59.2	68.3	76.6	83.6	90.1	93.4	91.2	86.2	72.5	63.5	58.2
		MCWB	54.2	51.2	54.6	58.8	61.8	66.5	68.0	67.3	64.7	58.4	57.9	54.4
	2%	DB	54.2	55.4	61.6	68.6	77.4	82.4	88.1	86.4	81.2	67.5	58.9	54.0
		MCWB	51.3	49.9	51.4	54.6	60.0	63.9	66.4	66.1	62.9	57.8	55.4	51.6
	5%	DB	51.7	53.1	57.4	63.6	72.5	77.1	83.6	82.3	77.0	63.6	55.6	51.0
		MCWB	49.3	48.4	49.9	52.3	58.1	61.3	65.3	64.7	62.0	56.2	53.1	49.2
	10%	DB	49.4	50.3	54.4	59.2	67.8	72.3	79.3	79.0	72.8	60.8	54.0	48.2
		MCWB	47.7	47.0	48.7	50.3	56.6	59.6	63.3	63.5	60.4	55.4	52.0	46.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	55.0	56.6	60.5	64.9	68.7	70.6	69.9	68.1	62.5	59.7	55.2
		MCDB	57.8	56.9	63.4	73.4	76.4	85.0	89.3	85.2	78.2	66.2	61.6	57.9
	2%	WB	52.1	51.9	53.7	56.4	61.8	65.3	68.0	67.7	65.4	59.9	56.2	52.0
		MCDB	53.1	53.4	58.7	64.9	72.8	79.5	84.6	82.4	76.3	63.9	58.2	53.4
	5%	WB	49.7	49.3	51.7	54.0	59.6	62.5	66.1	66.0	63.5	57.9	54.1	49.3
		MCDB	50.7	51.3	55.8	60.9	71.1	74.1	81.0	79.6	73.0	61.8	55.6	50.6
	10%	WB	47.4	47.2	49.8	51.9	57.3	60.4	64.3	64.5	61.9	56.1	52.1	47.0
		MCDB	48.6	49.7	53.7	57.7	66.3	70.5	77.3	76.6	70.6	60.0	53.4	47.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	15.1	17.2	19.0	20.1	20.3	23.4	23.6	22.4	17.5	14.2	11.5
		MCDBR	14.2	19.4	23.1	27.8	31.1	30.7	32.8	31.8	31.2	22.9	15.1	12.0
		MCWBR	10.9	13.1	14.1	15.0	15.5	14.0	13.9	13.9	15.6	14.1	11.5	9.8
	5% WB	MCDBR	12.1	14.3	19.1	23.1	26.1	26.3	28.6	27.5	25.5	17.8	13.0	11.2
		MCWBR	10.3	10.8	12.7	13.2	13.5	12.7	12.8	12.5	13.5	12.2	10.5	9.8
(40) Clear-Sky Solar Irradiance	taub		0.325	0.314	0.328	0.355	0.375	0.364	0.349	0.354	0.342	0.330	0.324	0.331
	taud		2.479	2.540	2.521	2.419	2.380	2.435	2.499	2.488	2.521	2.558	2.515	2.394
	Ebn at Noon		242	274	286	286	282	285	288	283	277	264	241	221
	Edh at Noon		21	24	28	34	37	35	32	31	28	23	20	20
(44) All-Sky Solar Radiation	RadAvg		326	596	885	1296	1642	1797	2008	1719	1265	718	383	269
(45)	RadStd		50	82	108	123	139	183	153	122	117	81	50	39

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+1.78	+2.00	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page